# Extraction and Mitigation of Radio Frequency Interference Artifacts Based on Time-Series Sentinel-1 SAR Data

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***Abstract*— Radio frequency interference (RFI) is a critical issue for accurate remote sensing by synthetic aperture radar (SAR). Existing literature mainly detects and mitigates RFI in the raw data domain, which is generally not accessible to the end-user. In this article, a novel RFI extraction and mitigation scheme in the image domain is proposed using multitemporal analysis of SAR images. By exploiting the coupling correlation and complementary information among the time-series images, the background landscape could be modeled as relatively stationary with the low-rank property. Meanwhile, the radiometric artifacts corresponding to RFI could be well extracted and characterized by the sparse components. Extraction and mitigation of RFI signatures could be achieved simultaneously via a joint iterative optimization process. Experimental results on typical real-measured Sentinel-1 datasets acquired in different regional areas with various RFI types demonstrate the validity of the proposed method.**

***Index Terms*— Multitemporal analysis, radio frequency interference (RFI), Sentinel-1, synthetic aperture radar (SAR).**

## I. Introduction

### A. Background and Motivation

SYNTHETIC aperture radar (SAR) is a powerful active remote sensing instrument, which could provide high-resolution images of land cover regardless of light and weather conditions [1]. With the proliferation of radio technology, many more radio services emerged and compete for access to the finite spectrum, making the electromagnetic environment getting more congested and contested [2]. The SAR system may collect spurious signals radiating from other transmitters operating within the same frequency or in nearby bands, such as wireless communication stations, ground-based radar systems, wind profilers, and even satellites. These spurious signals that appeared in the radar echoes are generally referred to as radio frequency interference (RFI) [3].

In recent years, many RFI cases have been observed in spaceborne SAR systems especially at lower frequencies, such as L- and C-bands, including PALSAR [4], RADARSAT and Sentinel-1 [5], Gaofen-3 [6], and so on. Fig. 1 illustrates the heatmap of RFI emitters spreading over the global area detected by the European C-band Sentinel-1 system. It is shown that the RFI situation differs significantly in geographical areas and is very severe in densely populated areas. Four representative datasets from different areas with typical RFI signatures are used for performance evaluation.

The presence of RFI in SAR echoes would produce adverse impacts in multiple aspects. First, it distorts the amplitude and phase correlation of the raw echoes, which would affect the image formation process and degrade the focusing quality of imaging results. Second, the artifacts superimposed on the image pose hindrance for the subsequent image interpretation, such as target detection and classification. Santamaria *et al.* [8] show that RFI can cause a lot of false alarms in the mass processing of Sentinel-1 images for ship detection, and part of the images are discarded directly. Furthermore, the accuracy of postprocessing products using interferometric and polarimetric information in quantitative remote sensing is also vulnerable to RFI [9]. Therefore, in order to preserve the valuable data, it is crucial to extract and mitigate the RFI signatures in SAR images.

### B. Related Works

A lot of efforts have been devoted to develop RFI detection and mitigation schemes [10]. The most practical signal processing method is the adaptive filtering method, which designs filters in time [11], or frequency [4], [12], or time–frequency domain [13]. It is simple and efficient, while it may confront relatively large signal loss when dealing with complex RFIs. More advanced techniques can be categorized into two classes, i.e., the model-based methods and the data-driven methods.

The *model-based methods* focus on using specific mathematical models to represent interference under the support of certain prior constraints and seek to decompose the echoes or reconstruct the interference under the optimization framework [14], [15]. Based on the hypothesis of joint low rank and sparse model, some researchers proposed a series of interference separation and signal reconstruction methods based on the robust principal component analysis [16]–[19]. From the perspective of sparse reconstruction theory, Joy *et al.* [20] proposed a semiparametric optimization

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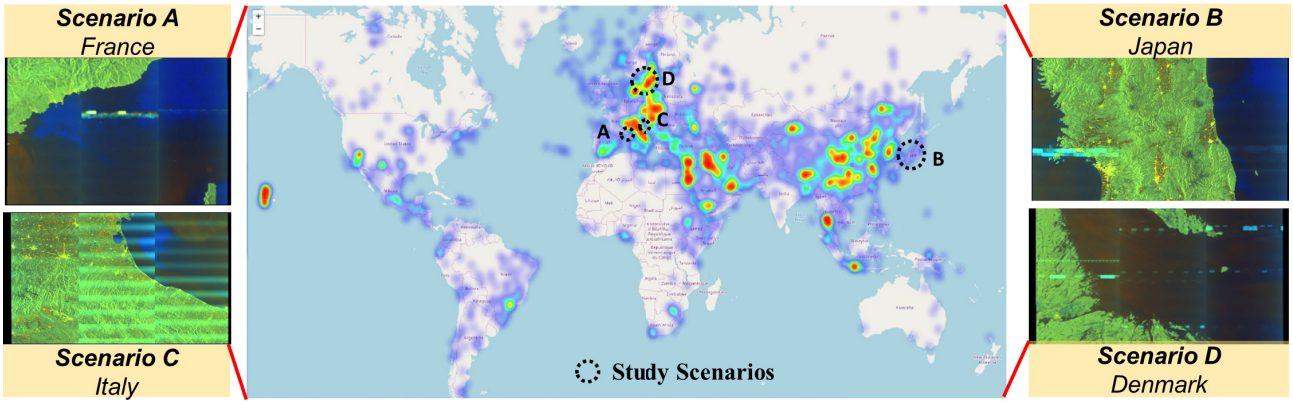



Fig. 1. Observations of RFI sources in the C-band spread over the global area detected by the European Sentinel-1 SAR system [6]. The dashed-circled region represents the dataset used for experiments.

framework. Furthermore, sparsity in range spectrum [20] and time–frequency domain [21] is investigated to suppress narrowband and wideband interferences. Considering the intra block correlation, the block sparse Bayesian learning shows the effectiveness of dealing with the highly underdetermined restoration problem [23]. Huang *et al.* [24] proposed a smoothing multiview tensor model for narrowband and wideband RFI mitigations. Generally, the performance advantage of the model-based method is directly determined by the selection of optimization models under specific prior knowledge constraints. The calculation of iterative optimization steps is cumbersome, and the effectiveness is difficult to be guaranteed since the selection of parameters of different datasets is not a simple task.

The other class of state-of-the-art techniques is the *data-driven method*. With the accumulation of big data of continuous imaging observation, as well as the advancement of deep learning methods, the RFI detection and mitigation methods are evolving to be more intelligent-oriented. Existing literature shows that the introduction of artificial intelligence technology, such as the deep learning network, can directly realize end-to-end learning without a traditional modeling process [24]–[26]. Through complex network structure and large-scale supervised training, the deep level intrinsic property differences between interference and echo could be captured, and the interference signatures can be intelligently eliminated. The essence of this model is a black-box model based on large samples, which requires a high quantity and quality of training data and lacks interpretability in the process.

For end-users of remote sensing, the level-0 raw echo is always not accessible and difficult to process. However, the majority of the mentioned methods are only capable of dealing with the raw echo data before focusing on processing, with few literature investigating the image-domain techniques for RFI detection and localization without considering RFI mitigation [27]–[29]. The investigated RFI types are limited to certain types, such as linear frequency modulated. Besides, these methods only utilize one single image and did not make full use of the time correlation between the revisit acquisitions of spaceborne SAR systems.

### C. Contributions

The objective of this article is to propose a novel RFI detection and mitigation scheme in the image domain based on multitemporal joint analysis as an alternative choice. The main contributions of this article are concluded as follows.

1) The interfering mechanism of terrestrial and terrain scattered RFI is investigated, and their radiometric characteristics in time-series images are analyzed. It provides a survey view of the current RFI situation in C-band spaceborne SAR, such as Sentinel-1.
2) It proposes an operational framework for joint RFI extraction and mitigation on Level-1 focused time-series images, which is applicable for both terrestrial RFI sources and terrain scattered mutual interference. To the best of our knowledge, this is the first published work that takes full advantage of the correlation and coupling among the time-series images for RFI mitigation.
3) Extensive experiments using Sentinel-1 data with various scenarios and RFI types are provided. It demonstrates that the proposed method can successfully extract the RFI signatures and mitigate the RFI impacts simultaneously with slight distortion to focused imaging results.

The rest of this article is organized as follows. Section II analyzes the interfering mechanism and analyzed the RFI characteristics among time-series images. Section III introduces the flowchart of the proposed method. Section IV provides extensive experimental results of Sentinel-1 datasets under various scenarios. Section V concludes this article.

## II. RFI Jamming Mechanism and Characteristics

Fig. 2 illustrates possible RFI sources and their interfering mechanisms for a spaceborne SAR system. Most of the RFI sources are associated with human activities over land. Terrestrial radio devices are considered as major contributors, including, but not limited to, telecommunication wireless communication stations, ground-based radiolocation radar, wind profilers, and so on, which have been widely reported and vary in signal characteristics and modulations [3].

Besides the terrestrial RFI sources, there are more and more case reports of mutual interference between satellites.

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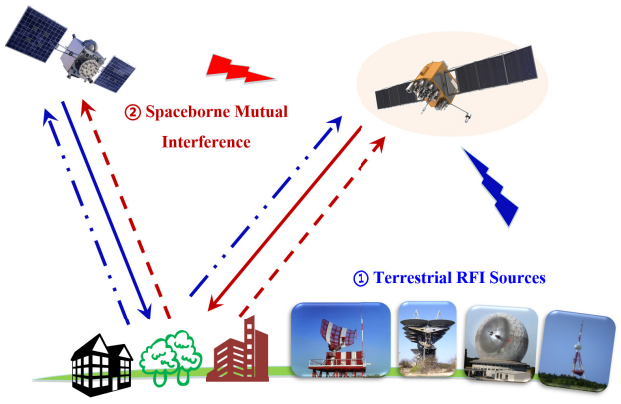



Fig. 2. Illustration of potential RFI sources for a spaceborne SAR system.

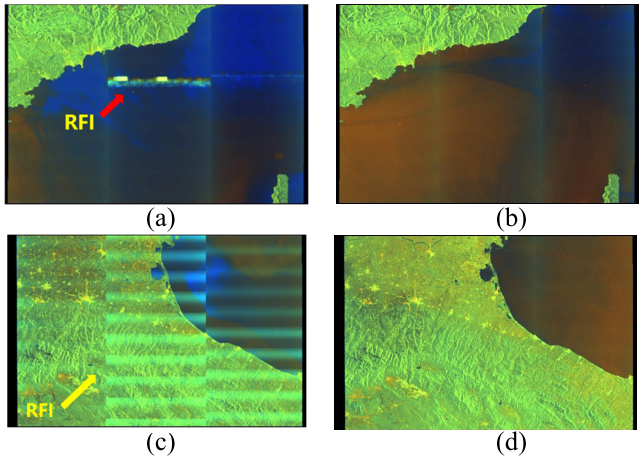



Fig. 3. Comparison of Sentinel-1 time-series images with and without (a) and (b) terrestrial RFI and (c) and (d) spaceborne mutual interference.

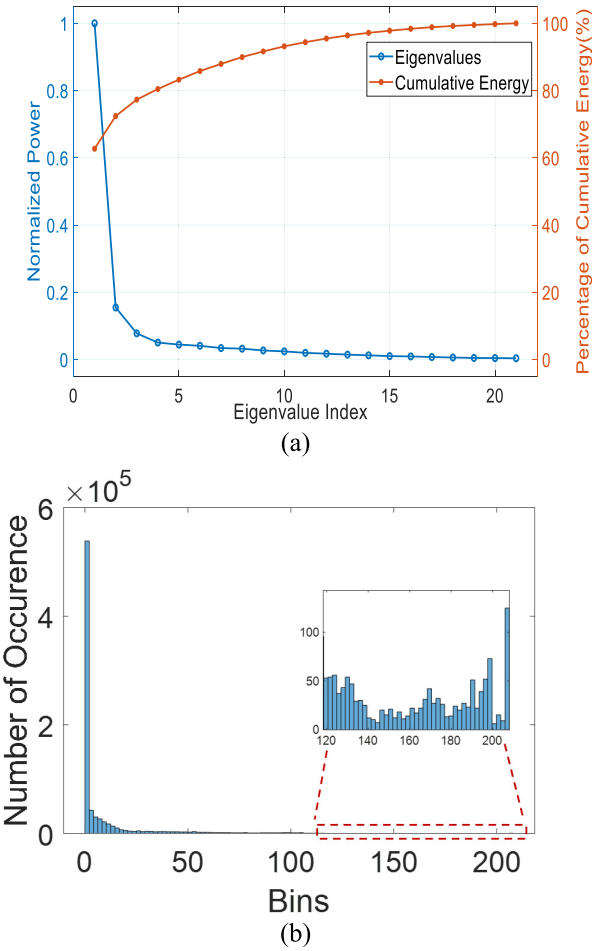



Fig. 4. Illustration of low-rank and sparsity property between time-series images. (a) Distribution of eigenvalues. (b) Sparsity level.

The mutual interference may come from direct emissions due to the footprint overlap in different altitudes and occur during the SAR data acquisition resulting from the signal backscattered from the illuminated surface [29]. This kind of mutual interference would be more common with the emergence of forthcoming SAR constellations and GEOSAR [30], [31].

Due to the different interfering mechanisms of the RFI sources, these two types of RFIs present unique radiometric characteristics on remote sensing images in terms of pattern, intensity, and size. The spaceborne SAR system has a fixed revisit duration for an orbit, for example, Sentinel-1A has a 12-day repeat cycle. Fig. 3 provides a comparison of the SAR images with and without the contamination of RFI acquired at two different passes. Compared with Fig. 3(b), Fig. 3(a) shows a scene with terrestrial RFI, in which a bright strip appeared on the center and presents unique radiometric property compared with the land cover and sea. In contrast to Fig. 3(d), Fig. 3(c) shows the image with spaceborne mutual RFI artifacts, where many strong stripes with high intensity appeared and produce irregular texture patterns. It severely distorts the image information and buries the true echoes of the illuminated area.

From Fig. 3, two important remarks could be summarized. The time-series images acquired at adjacent passes are quite similar in radiometric properties. This is because the land covers undergo slight variation in a relatively short time period. The time-series images are spatial–temporal correlated and could be expressed as a low-rank matrix. Fig. 4(a) shows the eigenvalues obtained by the eigenvalue decomposition for seven images. It is shown that the energy is predominated by the first few eigenvalues and decays significantly for the rest, which indicates a strong correlation and low-rank property. On the other hand, considering the nature of the interfering mechanism, the occurrence of interference is associated with several factors, including the signal characteristics of the RFI emitter, the configuration mode for a particular observation, and so on. Therefore, the RFI signatures in different acquisitions will be varying in spatial and pattern, and shows a certain degree of randomness and sparsity, as shown in Fig. 4(a). Seven time-series images of the same scenario are used for eigendecomposition. Each image is expanded into three column vectors (R, G, and B channels) and combined to get a two-dimensional matrix, which is used for eigendecomposition. Fig. 4(b) shows the histogram of the RFI residuals. The horizontal axis is the gray value of the image pixels, ranging from 0 to 255. The vertical axis is the cumulative pixels corresponding to different pixel values. It is shown that only a small portion of pixels dominates the energy contribution.

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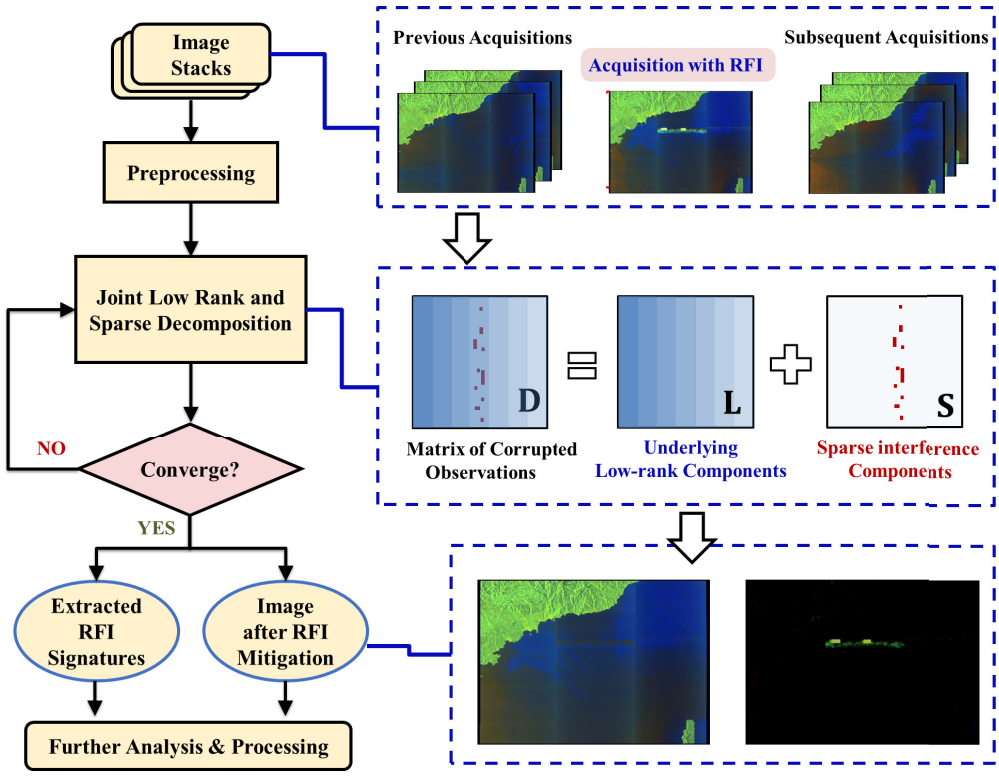



Fig. 5. Flowchart of the proposed RFI detection and mitigation scheme using time-series images.

## III. Theory and Methodology

Therefore, this work aims to develop a mathematical model that aims to exactly recover a low-rank matrix corrupted by sparse errors. More specifically, assume that a set of time-series images with $t = 1, \ldots, T$ is given, and stack them as vectors in a matrix. Then, according to the abovementioned remarks, the time-series images could be expressed as the following general model:

$$\mathbf{D} = \mathbf{H}_1\mathbf{L} + \mathbf{H}_2\mathbf{S} + \mathbf{N} \tag{1}$$

where $\mathbf{D} \in \mathbb{R}^{MN\times T}$ represents the data matrix, $M$ and $N$ denote the distance and azimuth dimension of SAR image, and $T$ denotes the number of images. $\mathbf{H}_1, \mathbf{H}_2$ are the measurement matrices. $\mathbf{L} \in \mathbb{R}^{MN\times T}$ is assumed to be a low-rank matrix corresponding to the stationary scene background, while $\mathbf{S} \in \mathbb{R}^{MN\times T}$ denotes the RFI signatures, which presents sparsity. $\mathbf{N} \in \mathbb{R}^{MN\times T}$ is the noise matrix.

In this model, the separation of RFI is essentially a joint optimization problem with the low-rank and sparsity constraints, which is formulated as

$$\min_{\mathbf{L},\ \mathbf{S}} \frac{1}{2}\|\mathbf{D} - (\mathbf{H}_1\mathbf{L} + \mathbf{H}_2\mathbf{S})\|_F^2 + \lambda_1\|\mathbf{L}\|_* + \lambda_2\|\mathbf{S}\|_{1,2} \tag{2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $\|\cdot\|_*$ denotes the nuclear norm, and $\|\cdot\|_{1,2}$ denotes the mixed $l_{1,2}$ norm. $\lambda_1$ and $\lambda_2$ are regularized hyperparameters.

The optimization problem in (2) could be further simplified as

$$\min_{\mathbf{L},\mathbf{s}} \frac{1}{2}\|\mathbf{D} - \mathbf{A}\mathbf{X}\|_F^2 + h(\mathbf{X}) \tag{3}$$

where $\mathbf{A} = [\mathbf{H}_1, \mathbf{H}_2]$, $\mathbf{X} = [\mathbf{L}; \mathbf{S}]$, $\mathbf{P}_1 = [\mathbf{I}; \mathbf{0}]$, $\mathbf{P}_2 = [\mathbf{0}; \mathbf{I}]$. $h(\mathbf{X}) = \sum_{i=1}^{2} \lambda_i\rho_i(\mathbf{P_i X})$, $\rho_1 = \|\cdot\|_*$, and $\rho_2 = \|\cdot\|_{1,2}$.

The minimization problem is a regularized least-squares problem and can be solved using the iterative shrinkage/thresholding algorithm by finding the proximal mapping [32]

$$\text{prox}_h(\mathbf{X}) = \arg\min\left\{\lambda_1\rho_1(\mathbf{D}) + \frac{1}{2}\|\mathbf{D} - \mathbf{L}\|_F^2 + \lambda_2\rho_2(\mathbf{D}) + \frac{1}{2}\|\mathbf{D} - \mathbf{S}\|_F^2\right\}. \tag{4}$$

With the separable property of $\text{prox}_h(\mathbf{X})$, it leads to

$$\text{prox}_h(\mathbf{X}) = \begin{bmatrix} \text{prox}_{\rho_1}(\mathbf{L}) \\ \text{prox}_{\rho_2}(\mathbf{S}) \end{bmatrix} = \begin{bmatrix} \text{SVT}_{\lambda_1}(\mathbf{L}) \\ \Gamma_{\lambda_2}(\mathbf{S}) \end{bmatrix} \tag{5}$$

where $\text{SVT}_\alpha(\mathbf{X}) = \mathbf{U}\text{diag}(\max(0, \sigma_i - \alpha))\mathbf{V}^H, i = 1, \ldots, r$ denotes the singular value thresholding operator, and $\Gamma_{\lambda_2}(\mathbf{S})$ denotes the soft-thresholding operator.

Then, the proximal mapping is applied to the gradient of the quadratic part in (3), which is given by

$$g(\mathbf{X}) = \frac{d}{d\mathbf{X}}\|\mathbf{D} - \mathbf{A}\mathbf{X}\|_F^2 = \begin{bmatrix} \mathbf{H}_1^H(\mathbf{H}_1\mathbf{L} + \mathbf{H}_2\mathbf{S} - \mathbf{D}) \\ \mathbf{H}_2^H(\mathbf{H}_1\mathbf{L} + \mathbf{H}_2\mathbf{S} - \mathbf{D}) \end{bmatrix} \tag{6}$$

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where $g(\cdot)$ denotes the gradient of the quadratic term. Therefore, the iterative step for solving (3) is given by

$$\mathbf{X}^{k+1} = \text{prox}_h\left\{\mathbf{X}^k - \frac{1}{L_f} g\left(\mathbf{X}^k\right)\right\}. \tag{7}$$

Equivalently, it could also be expressed as

$$\mathbf{L}^{k+1} = \text{SVT}_{\lambda_1/L_f}\left\{\left(\mathbf{I} - \mathbf{H}_1^H\mathbf{H}_1/L_f\right)\mathbf{L}^k - \mathbf{H}_1^H\mathbf{H}_2\mathbf{S}^k + \mathbf{H}_1^H\mathbf{D}\right\} \tag{8}$$

$$\mathbf{S}^{k+1} = \Gamma_{\lambda_2/L_f}\left\{\left(\mathbf{I} - \mathbf{H}_2^H\mathbf{H}_2/L_f\right)\mathbf{S}^k - \mathbf{H}_2^H\mathbf{H}_1\mathbf{L}^k + \mathbf{H}_2^H\mathbf{D}\right\} \tag{9}$$

where $L_f$ is the Lipschitz constant of the quadratic term of (3), and $\mathbf{L}^{k+1}$ and $\mathbf{S}^{k+1}$ are the low-rank component and the sparse component after $k$ times iteration. The iterative process will not be terminated until reaching the maximum iterations or acceptable estimation error.

Fig. 5 summarizes the flowchart of the proposed RFI detection and mitigation scheme using time-series images. The RFI artifacts could be eliminated by decomposition of the time-series images under the joint low-rank and sparse constraints. The extracted RFI signatures could be useful for further connection with the tracking back analysis of radiation sources.

## IV. Experimental Results

In this section, experimental results of several real measured datasets from Sentinel-1A are provided to validate the effectiveness of the proposed method. As illustrated in Fig. 1, totally four datasets are taken from representative scenarios around the globe with various land cover types and differ for obvious RFI signatures, including the terrestrial RFI and spaceborne mutual RFI. The scenes are becoming more and more complicated with the expansion of the RFI contamination ratio, increasing the difficulty for RFI detection and mitigation.

If the number of image pairs is limited, the low-rank and sparse constraints are not fully met. Therefore, the redundancy and coupling among images are not sufficient to produce rather satisfactory recovery results. On the other hand, the mitigation performance of RFI will be improved as the number of images increases while with the further increase in the computational complexity due to the SVD computation process. However, according to our extensive study, the further performance improvement is slight with a large number of image pairs. Therefore, in our work, we have set the image number as 7 for experimental comparisons. The optimal values of $\lambda_1$ and $\lambda_2$ are derived from a great deal of experimental results.

### *A. Results of Scenario a in France*

The first dataset is acquired by Sentinel-1A near the coastline of France on July 23, 2020, as illustrated in Fig. 6(a). The size of the image is $416 \times 630$. Fig. 6(b)–(g) compares the six images acquired in different passes before and after the date. Sentinel-1 acquires the data by using the Terrain Observation with Progressive Scans SAR (TOPSAR) mode. The TOPSAR technique is a form of ScanSAR imaging, where data are acquired in bursts by cyclically switching the antenna beam between multiple adjacent subswaths. The vertical lines in

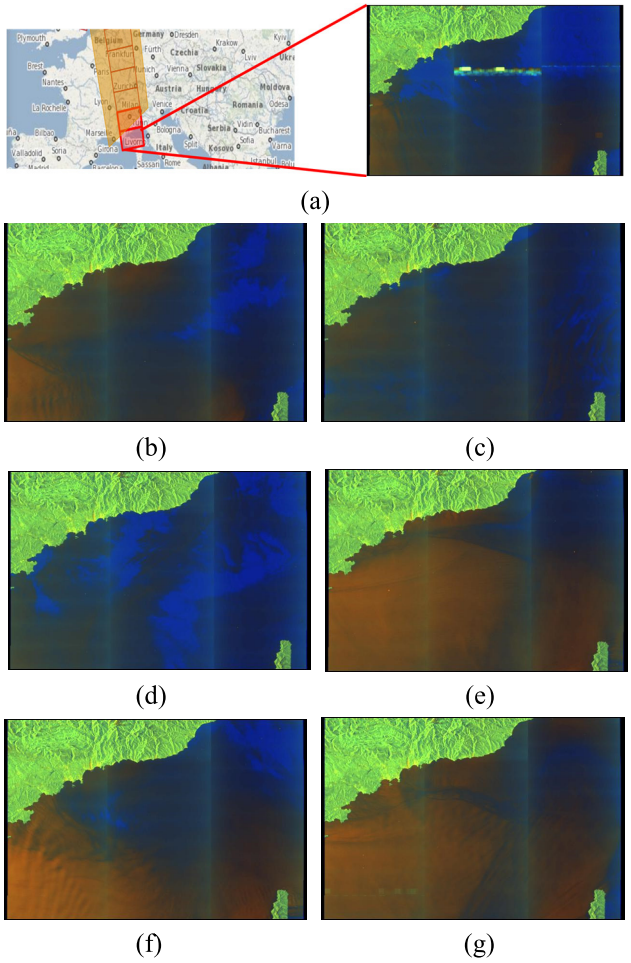



Fig. 6. Illustration of Scenario A in the France region. (a) Geographic location of the dataset and its corresponding SAR image with RFI artifacts. (b)–(g) Images without RFI artifacts acquired in adjacent passes.

Fig. 6 are a result of the stitching of the subswath data. From the time-series images, it is shown that there are strong abnormal artifacts appeared on the sea. The interference is a typical terrestrial RFI, which may originate from the ship's radio equipment traveling on the sea. Due to different sea states for each acquisition, the scattering coefficients of the sea surface have significant visual variation. This would degrade the spatial–temporal correlation of time-series images and increase the difficulty for effective elimination.

Fig. 7 shows the results of dataset A after applying the proposed scheme. Fig. 7(a) shows the sparse components that correspond to the RFI signatures. It is shown that the pattern and shape are effectively extracted, without introducing outliers that belong to the actual land cover and sea surface. Fig. 7(b) plots the extracted RFI signatures after binary thresholding, which is beneficial for the investigation of the RFI sources, i.e., to build a mapping connection with the signal characteristics and the resulting image artifacts. Fig. 7(c) plots the recovered image after RFI mitigation. It is shown that the contaminated image is well recovered, in which the RFI artifacts are eliminated without introducing many distortions to the image quality.

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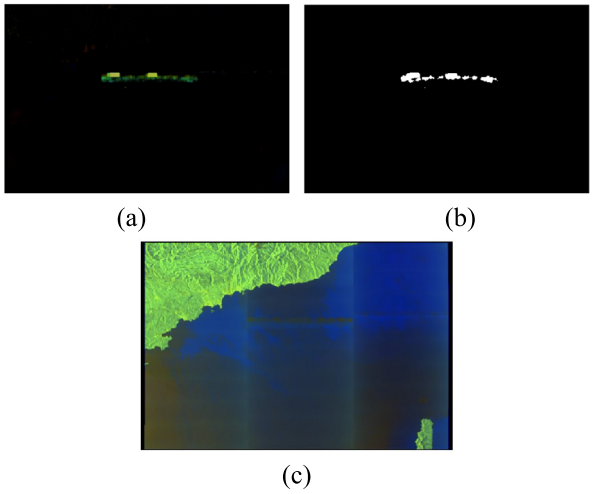



Fig. 7. Results of Scenario A after applying the proposed scheme. (a) Extracted RFI signatures. (b) RFI mask after binarization thresholding. (c) Recovered image after RFI mitigation.

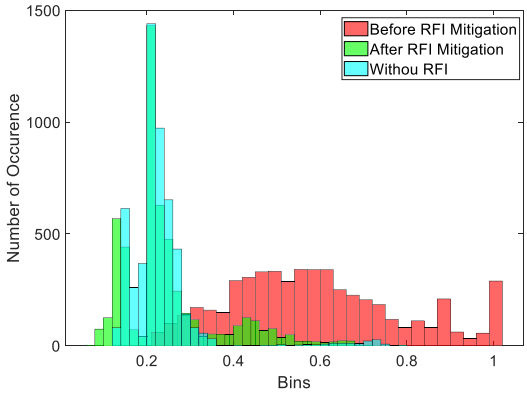



Fig. 8. Histograms before and after RFI extraction and mitigation for Scenario A.

Furthermore, Fig. 8 compares the partial histograms of the extracted RFI artifacts and the recovered image. It is shown that the histogram after RFI mitigation and the one without RFI are overlapped and very similar, while it differs significantly from that before RFI mitigation, indicating the effectiveness of the proposed scheme.

### *B. Results of Scenario B in Japan*

This second RFI-contaminated dataset is acquired by Sentinel-1A in the northeast region of Japan on April 10, 2021, as illustrated in Fig. 9(a). Fig. 9(b)–(g) compares the six images that are acquired in adjacent passes. This case is more complicated than Scenario A. The strong RFI blocks in the left-hand side overlapped with the continent, while weak signatures existed in the right part of the scene. The dynamic contrast between strong and weak interference in the same image increases the difficulty for effective extraction and mitigation.

Fig. 10 shows the recovered results after applying the proposed scheme. Under the sparsity regularization, the RFI signatures are extracted out together with some residuals from the land covers. The unexpected residuals can be further eliminated by binarization thresholding, as shown in Fig. 10(b).

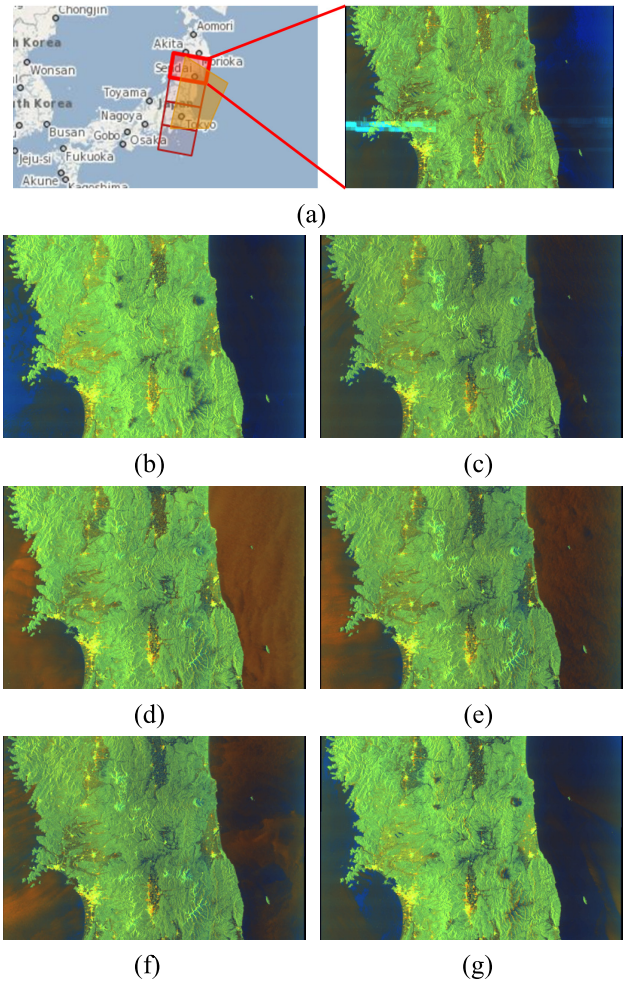



Fig. 9. Illustration of Scenario B in France area. (a) Geographic location of the dataset and its corresponding SAR image with RFI artifacts. (b)–(g) Images without RFI artifacts acquired in adjacent passes.

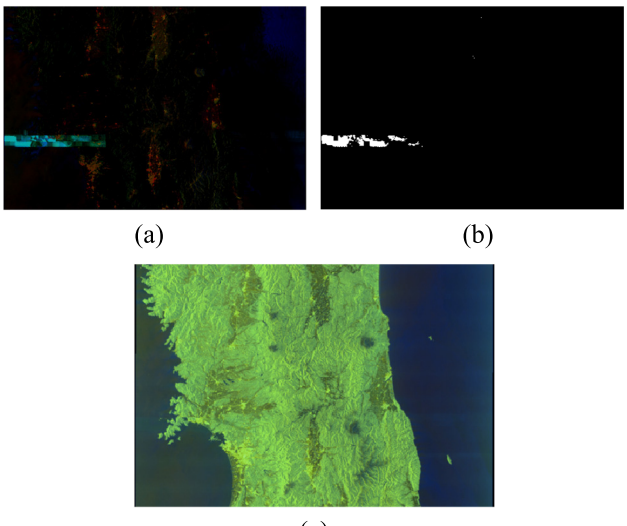



Fig. 10. Results of Scenario B after applying the proposed scheme. (a) Extracted RFI signatures. (b) RFI mask after binarization thresholding. (c) Recovered image after RFI mitigation.

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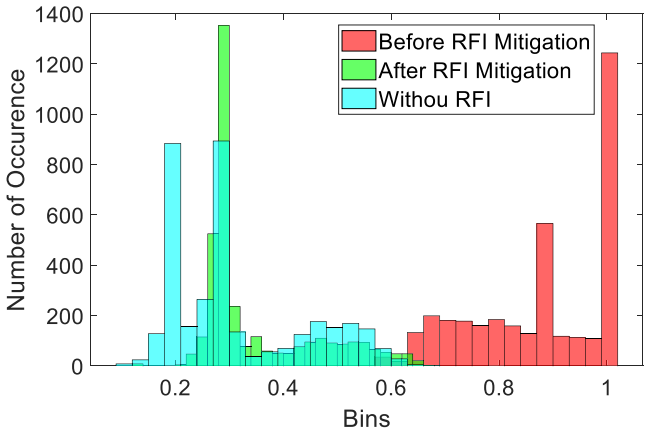



Fig. 11. Histograms before and after RFI extraction and mitigation for Scenario B.

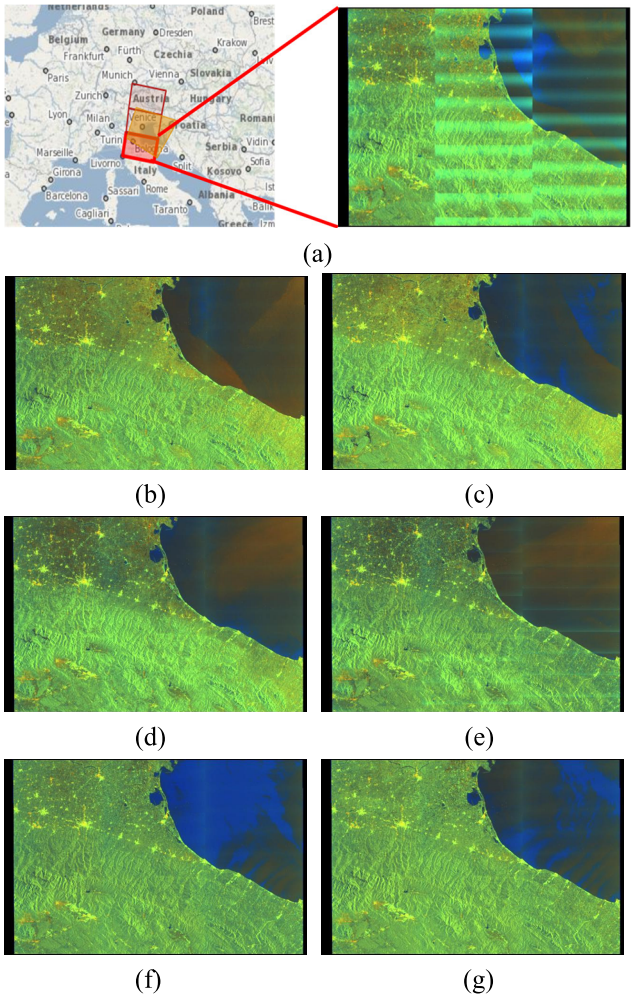



Fig. 12. Illustration of Scenario C in northern Italy. (a) Geographic location of the dataset and its corresponding SAR image with RFI artifacts. (b)–(g) Images without RFI artifacts acquired at different acquisitions.

Fig. 10(c) shows the recovered image after RFI mitigation. It is shown that both the strong RFI artifacts on the left-hand side and weak RFI signatures are eliminated out and preserve the details of the land cover well. Furthermore, Fig. 11 compares

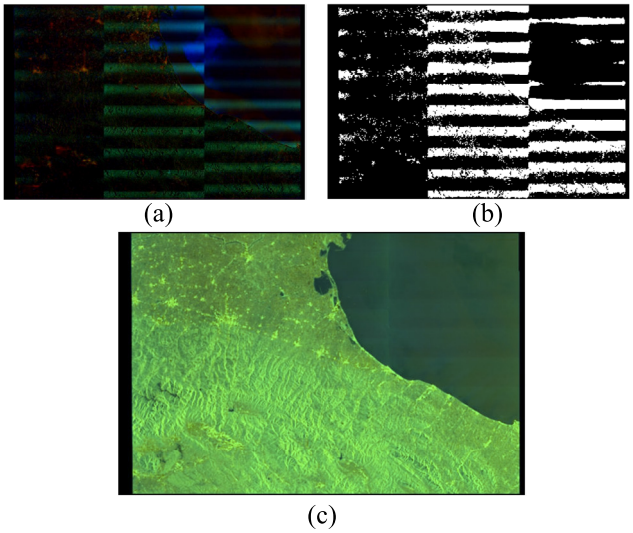



Fig. 13. Results of Scenario C after applying the proposed scheme. (a) Extracted RFI signatures. (b) RFI mask after binarization thresholding. (c) Recovered image after RFI mitigation for Fig. 12(a).

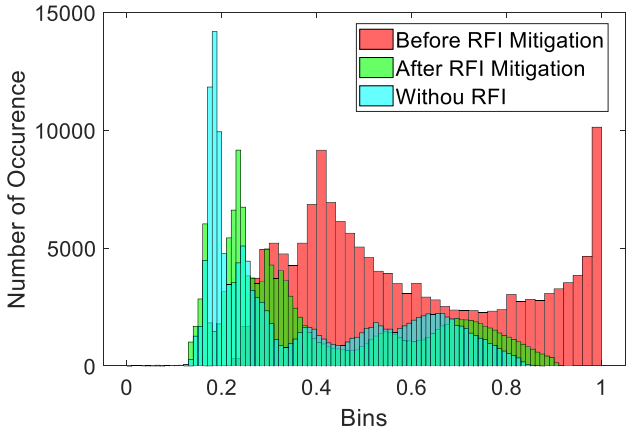



Fig. 14. Histograms before and after RFI extraction and mitigation for Scenario C.




Fig. 15. Geographic illustration of Scenario D for study.

the histograms before and after RFI extraction and mitigation for Scenario B. The distribution similarity in shape and magnitude indicates the effectiveness of the proposed scheme.

### C. Results of Scenario C in Italy

The third dataset is acquired by Sentinel-1A in the northern region of Italy on April 14, 2021, as illustrated in Fig. 12(a). The size of the image is 411 × 638. Fig. 12(b)–(g) compares

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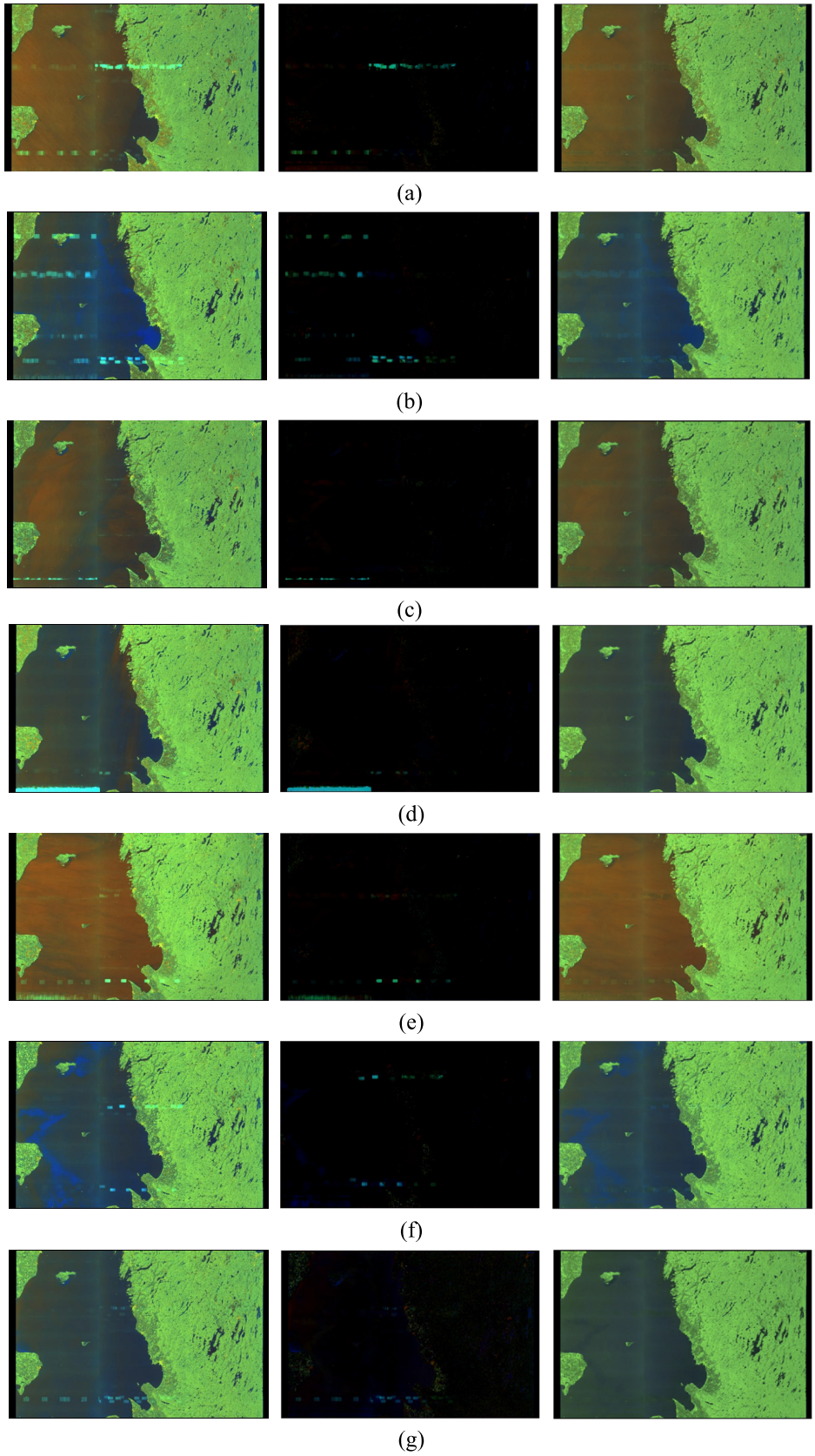



Fig. 16. Experimental results of scenario D. (a)–(g) Processing results for different acquisitions. The first column is the original RFI-contaminated image. The second column represents the extracted RFI signature by the proposed method, and the third column represents the corresponding images after RFI mitigation.

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the six images that are acquired in adjacent passes at different revisit times. The RFI case in this scenario is a typical spaceborne mutual interference. It is straightforward to notice that many periodic bright stripes with varying intensity appeared to present unique texture patterns. This is a very special type of mutual interference from other remote sensing satellites operating in the same frequency. Other SAR systems also transmit LFM signals, which possess certain coherence with the emitted signal with that of Sentinel-1. Since Sentinel-1 acquired the data in TOPSAR mode, the long duration mutual RFI exist in most bursts. When the radar receives the signal, the interference signal will be superimposed on the echo signal. After imaging, the intensity of regular stripes is dominated by many range lines with high intensity, which is a result of partial coherent processing.

Compared with the RFI in Scenario A and Scenario B with good energy concentration (as shown in Figs. 6 and 9), the energy of the terrain-scattered interference is spanned to cover the majority part of the image, which introduces severer distortion to the measurements and increases the difficulty for RFI extraction and mitigation greatly.

Fig. 13 shows the results of Scenario C after applying the proposed scheme. From Fig. 13(a) and (b), the extracted RFI signatures clearly present the periodic radiometric pattern caused by terrain scattered mutual interference. It is also shown that many isolated points are also included. This phenomenon is caused by the dispersing energy property of the terrain scattered interference, which weakens the sparsity constraints in the assumption model. Meanwhile, for a single pixel, the signature is a coherent mixture of RFI and target response, which is hard to clearly distinguish the actual contribution. However, as shown in Fig. 13(c), the quality of the recovered image after RFI mitigation is rather well, in which the mountain texture and contour of land cover are sufficiently clear, so that the subsequent image interpretation process is not affected. The comparison of histograms in Fig. 14 also verifies the validity.

### *D. Results of Scenario D in Denmark*

This dataset is acquired by Sentinel-1A in the Baltic Sea near Denmark from April 5 to July 22, 2021, as shown in Fig. 15. Previous scenarios include various types of RFI signatures in a single or two images. This dataset is much more complicated, in which the RFI artifacts present, in every acquired image, various spatial locations, and radiometric patterns, as shown in the first column of Fig. 16. Fig. 16(a)–(g) compares the seven images acquired in adjacent passes at different revisit times. As shown in Fig. 16, the pattern and location of the interference change at different times, which indicates that it is not a stationary RFI source. In addition, as illustrated in Fig. 15, the dataset of Scenario D is taken in the vicinity of the Baltic Sea near Denmark, which is a very busy international shipping lane. Therefore, it is inspected that there is a great possibility that the RFI may originate from the radiolocation equipment, such as radars mounted on ships.

After applying the proposed method, the extracted RFI signature corresponding to each image acquisition is shown in the second column. It is shown that the pattern of the RFI signatures is clearly extracted out with few noise outliers. Although the RFI artifacts span along with all the acquisitions, it still obeys the sparsity constraint with a small occupying percentage of the image pixels. The changing geometry of the observation and the time-varying property of the RFI produce artifacts in different spatial locations. The third column represents the corresponding image after RFI mitigation. It is shown that all the images are well recovered, and the hidden land covers are revealed by mitigating the RFI signatures. Fig. 17 plots the converge curve of the optimization process. It indicates that the proposed method is quite efficient and could achieve an optimal solution within a small number of iterations.

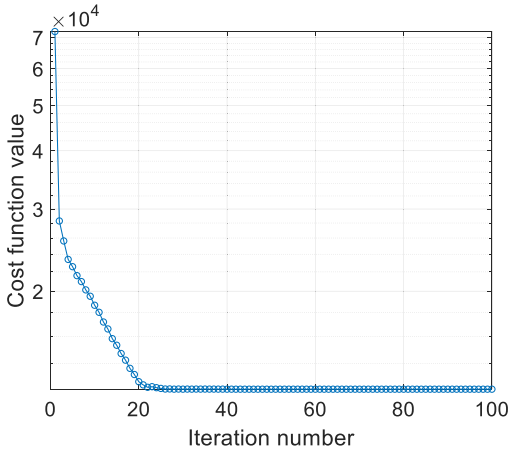



Fig. 17. Convergence of the optimization process.

## V. Conclusion

RFI detection and mitigation in the SAR image have been a hot topic and also a difficult task. Most of the existing works mainly deal with Level-0 raw data while seldom paying attention to processing techniques on the Level-1 image data. The Level-0 raw echo is often not accessible and difficult to process for end-users. In this article, we proposed a novel joint RFI extraction and mitigation scheme in the image domain based on the coupling analysis of time-series images. Its merits lie on the following:

1) The proposed method takes advantage of the spatial–temporal correlation of the land covers, as well as the time-varying property of the RFI artifacts. These fundamental assumptions lead to a joint low-rank and sparse model optimized for RFI extraction and mitigation.
2) It is capable of dealing with various RFI types, ranging from typical terrestrial RFI and complex mutual terrain-scattered interference to time-varying interferences spanning through multiple acquisitions. Extensive experimental results on several diverse scenarios acquired by Sentinel-1A are demonstrated.
3) It verifies that the proposed method can successfully extract the RFI signatures and mitigate the RFI impacts without introducing large distortion to focused imaging

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results. The extracted RFI signatures are beneficial for further investigation of the possible radiation sources and RFI case reporting.

4) The level-1 quick-look data are used to verify the feasibility of the proposed method. Similarly, this algorithm is also applicable to SLC and GRD data. However, these two types of data have larger image matrices, and the involving SVD decomposition has a large computation burden. Therefore, block partition and parallel processing are needed to improve efficiency. It is worth noting that the RFI postprocessing is not the best way to tackle the RFI issue, and there is no universal common strategy for mitigating the RFI impacts. The mitigation always is realized with a certain degree of signal distortion. The proposed method provides a complementary alternative to existing techniques, which is quite suitable for mass processing applications.

However, the hyperparameter is selected empirically, which cannot be adjusted adaptively for different scenarios. In future work, we would like to investigate a model-based learning strategy to adaptively learn the hyperparameters.

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